

STATISTICS

Regression and Correlation Analysis

PART 2: Estimation and Hypothesis for Regression Parameters

Birol Aslanyürek

Rev: 26.12.2023

Point Estimation of Regression Parameters

When we select a sample and then obtain a regression line

$$\hat{Y} = b_0 + b_1 X,$$

the following estimations can be made:

$$\beta_0 = b_0$$

$$\beta_1 = b_1$$

$$\rho = r$$

$$Y_k = \hat{Y}_k \text{ for a given } x_k$$

$$\bar{Y}_k = \hat{Y}_k \text{ for a given } x_k$$



point estimations

The above regression parameters can also be estimated by a confidence interval and/or be tested by hypothesis testing.

Correlation Coefficient ρ

For the correlation coefficient of the population ρ and its estimator r , the variable

$$t_{n-2} = \frac{r - \rho}{S_r}$$

t distribution with freedom degree $n-2$. Here n is the size of the sample and

$$S_r = \sqrt{\frac{1 - r^2}{n - 2}}$$

Is the standard deviation of the estimator r . S_r is also known as the **standard error of the correlation coefficient**.

Correlation Coefficient ρ

Confidence Interval for ρ :

The confidence interval with significance level α can be derived as:

$$P \left[-t_{\alpha/2, n-2} \leq \frac{r - \rho}{S_R} \leq t_{\alpha/2, n-2} \right] = 1 - \alpha$$



$$P \left[r - t_{\alpha/2, n-2} S_R \leq \rho \leq r + t_{\alpha/2, n-2} S_R \right] = 1 - \alpha$$



$$\left[r - t_{\alpha/2, n-2} S_R, \quad r + t_{\alpha/2, n-2} S_R \right]$$

Correlation Coefficient ρ

Hypothesis Testing for ρ :

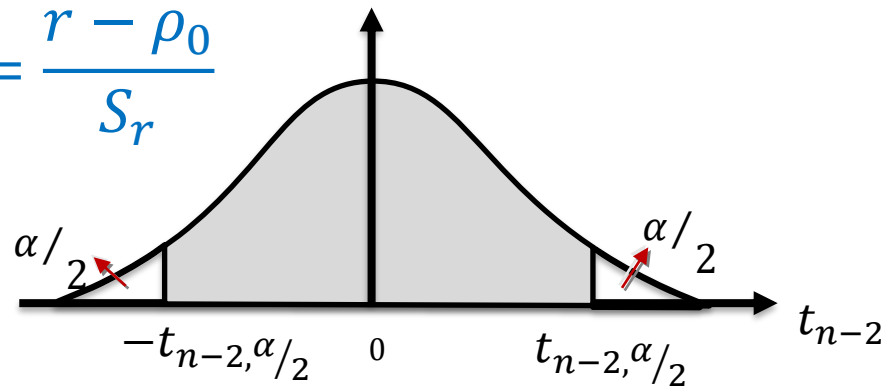
While it is possible to test one-tailed and two-tailed hypotheses, the most commonly used hypotheses are:

$H_0 : \rho = 0 (\rho_0)$ (There is no correlation.)

$H_a : \rho \neq 0 (\rho_0)$ (There is a correlation.)

The test statistic is: $t_0 = \frac{r - \rho_0}{S_r}$

The decision rule:



Accept H_0 , if $-t_{n-2, \alpha/2} \leq t_0 \leq t_{n-2, \alpha/2}$.

If we reject H_0 , it means that there is a significant relation between X and Y . If there is no logical relation between variables, the correlation is said to be fake (**correlation does not imply causation**).

Example: In a duration of one hour, the number of customers coming to a shop and their spending are given below.

The number of customers	25	20	50	35	40
Spending	12.5	10.4	25.3	20.2	24.1
(x 10000 TL)					

If we test the correlation coefficient at 5% importance level, test hypotheses are

$$H_0 : \rho = 0$$

$$t_0 = \frac{r - \rho_0}{S_r}$$

$$H_1 : \rho \neq 0$$

$$n - 2 = 5 - 2 = 3$$

$$\alpha/2 = 0.025$$

$$t_{n-2, \alpha/2} = 3.182$$

If we assume that the independent variable X is the number of customers and the dependent variable Y is the amount of spending, the correlation coefficient is;

$$r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

	X	Y	XY	X ²	Y ²
	25	12.5	312.5	625	156.2
	20	10.4	208	400	108.1
	50	25.3	1265	2500	640.09
	35	20.2	707	1225	408.04
	40	24.1	964	1600	580.81
Total	170	92.5	3456.5	6350	1893.3

$$r = \frac{5(3456.5) - 170(92.5)}{\sqrt{[5(6350 - 170^2)][5(1893.3) - (92.5)^2]}} = 0.9669$$

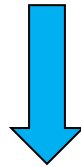
$$t_0 = \frac{r - \rho_0}{\sqrt{\frac{1 - r^2}{n - 2}}} = \frac{0.9669 - 0}{\sqrt{\frac{1 - (0.9669)^2}{5 - 2}}} = 6.5635$$

$$t_{n-2, \alpha/2} = 3.182$$

$$t_0 = 6.5635$$



$$t_0 > t_{n-2, \alpha/2}$$



We reject hypothesis H_0 for the 5% importance level. There is a correlation between variables.

Population Parameters β_0 and β_1

The means and variances of the statistics b_0 and b_1 are:

$$\begin{aligned} E(b_0) &= \beta_0 & \text{Var}(b_0) &= \frac{\sum X^2}{\sum X^2 - (\sum X)^2/n} \sigma^2 \\ E(b_1) &= \beta_1 & \text{Var}(b_1) &= \frac{1}{n[\sum X^2 - (\sum X)^2/n]} \sigma^2 \end{aligned}$$

where σ^2 is the variance of Y .

If the statistics b_0 and b_1 are distributed normally, the variables:

$$z = \frac{b_0 - \beta_0}{\sqrt{\text{Var}(b_0)}}$$

$$z = \frac{b_1 - \beta_1}{\sqrt{\text{Var}(b_1)}}$$

have normal distributions.

If we do not know the population parameter σ^2 , instead of it we can use its estimator

$$s^2 = \frac{\sum (Y - \hat{Y})^2}{n - 2} = \frac{\sum e^2}{n - 2}$$

which is called the variance of estimation (estimator). Here, S is the standard error of estimation (estimator).

In this case, the variables distributed normally are converted into:

$$t_{n-2} = \frac{b_0 - \beta_0}{\sqrt{\text{Var}(b_0)}} \quad t_{n-2} = \frac{b_1 - \beta_1}{\sqrt{\text{Var}(b_1)}}$$

where

$$\text{Var}(b_0) = \frac{\sum X^2}{\sum X^2 - (\sum X)^2/n} s^2 \quad \text{Var}(b_1) = \frac{1}{n[\sum X^2 - (\sum X)^2/n]} s^2$$

Population Parameters β_0 and β_1

Confidence Interval for β_0 and β_1 :

The confidence interval with significance level α can be derived as:

$$P \left[-t_{\alpha/2, n-2} \leq \frac{b_0 - \beta_0}{\sqrt{\text{Var}(b_0)}} \leq t_{\alpha/2, n-2} \right] = 1 - \alpha$$



$$P \left[b_0 - t_{\alpha/2, n-2} \sqrt{\text{Var}(b_0)} \leq \beta_0 \leq b_0 + t_{\alpha/2, n-2} \sqrt{\text{Var}(b_0)} \right] = 1 - \alpha$$



$$\left[b_0 - t_{\alpha/2, n-2} \sqrt{\text{Var}(b_0)}, \quad b_0 + t_{\alpha/2, n-2} \sqrt{\text{Var}(b_0)} \right]$$

Similarly for the other parameter:

$$\left[b_1 - t_{\alpha/2, n-2} \sqrt{\text{Var}(b_1)}, \quad b_1 + t_{\alpha/2, n-2} \sqrt{\text{Var}(b_1)} \right]$$

Population Parameters β_0 and β_1

Hypothesis Testing for β_0 and β_1 :

The most commonly used hypotheses for β_0 and β_1 are:

$$\begin{array}{l} H_0 : \beta_0 = a \\ H_a : \beta_0 \neq a \end{array} \quad \longrightarrow \quad \text{test statistic: } t_0 = \frac{b_0 - \beta_0}{\sqrt{\text{Var}(b_0)}}, (\beta_0 = a)$$

$$\begin{array}{l} H_0 : \beta_1 = b \\ H_a : \beta_1 \neq b \end{array} \quad \longrightarrow \quad \text{test statistic: } t_1 = \frac{b_1 - \beta_1}{\sqrt{\text{Var}(b_1)}}, (\beta_1 = b)$$

If $-t_{n-2, \alpha/2} \leq t_0$ (or t_1) $\leq t_{n-2, \alpha/2}$, we accept H_0 .

$H_0 : \beta_1 = 0$ means that β_1 is not significant statistically or there is **no linear relationship** between X and Y . (Y may be dependent on X , but not linearly.)

$H_0 : \beta_0 = 0$ means that β_0 is not significant statistically.

The Estimation of Y_k for a Specific Value of X_k

One of the most important purposes of regression analysis is to estimate the value of Y_k for a value of X_k .

$$\hat{Y}_k = b_0 + b_1 X_k$$

Here, $\hat{Y}_k$ is an estimator of the population parameter Y_k . The variable

$$t_{n-2} = \frac{\hat{Y}_k - Y_k}{S_{\hat{Y}_k}}$$

t distribution with freedom degree $r-2$. Here n is the size of the sample and the standard error of the estimation is

$$S_{\hat{Y}_k} = \sqrt{1 + \frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}} S$$

The Estimation of Y_k for a Specific Value of X_k

Note that if we interested in the **mean of Y_k** for a specific value of X_k in the population, the standard error of the estimation becomes

$$S_{\widehat{\mu_{Y_k}}} = \sqrt{\frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}} S$$

Y_k for a given x_k  $S_{\widehat{Y_k}} = \sqrt{1 + \frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}} S$

For example, the maintenance spending of a specific truck whose age is 4 years...

$\bar{Y}_k$ for a given x_k  $S_{\widehat{\mu_{Y_k}}} = \sqrt{\frac{1}{n} + \frac{(X_k - \bar{X})^2}{\sum X^2 - n\bar{X}^2}} S$

For example, the mean of the maintenance spending of trucks whose ages are 4 years...

The Estimation of Y_k for a Specific Value of X_k

Confidence Interval for Y_k :

The confidence interval with significance level α can be derived as:

$$P \left[-t_{\alpha/2, n-2} \leq \frac{\hat{Y}_k - Y_k}{S_{\hat{Y}_k}} \leq t_{\alpha/2, n-2} \right] = 1 - \alpha$$



$$P \left[\hat{Y}_k - t_{\alpha/2, n-2} S_{\hat{Y}_k} \leq Y_k \leq \hat{Y}_k + t_{\alpha/2, n-2} S_{\hat{Y}_k} \right] = 1 - \alpha$$



$$\left[\hat{Y}_k - t_{\alpha/2, n-2} S_{\hat{Y}_k}, \quad \hat{Y}_k + t_{\alpha/2, n-2} S_{\hat{Y}_k} \right]$$

ANOVA: Population Parameters β_0 and β_1

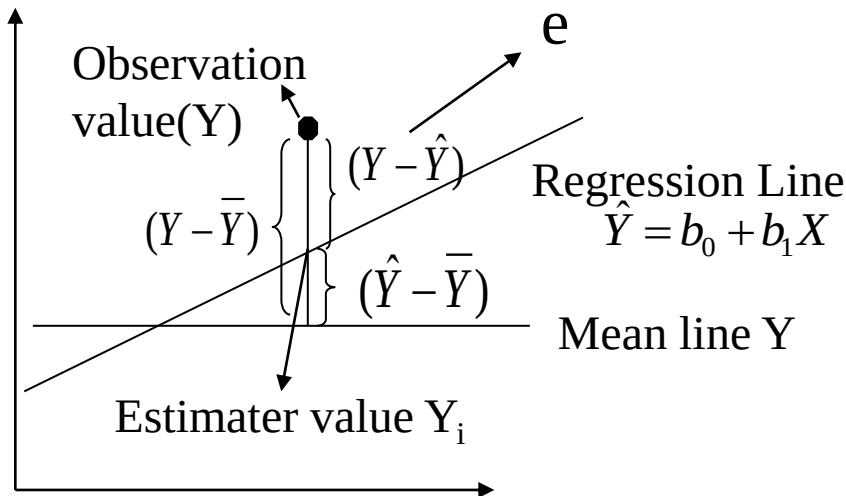
$$\sum (Y - \bar{Y})^2 = \sum (\hat{Y} - \bar{Y})^2 + \sum (Y - \hat{Y})^2$$

Sum of Total Squares (SST) Sum of Regression Squares (SSR) Sum of Error Squares (SSE)

$$SST = SSR + SSE$$

Thus, the coefficient of determination:

$$R^2 = \frac{SSR}{SST}$$



ANOVA Table for β_0 and β_1

The population parameters β_0 and β_1 can be tested separately as given before. However, they also can be tested by using a model containing both parameters at the same time. Accordingly, the hypothesis test can be applied using the table given below:

Variance Source	Square Sum	Degree of Freedom	Average Squares	The Proportion F
Regression	SSR	1	SRS/1	$F_{1,n-2,\alpha} = \frac{SRS}{\frac{SES}{n-2}}$
Error	SSE	n-2	SES/(n-2)	
Total	SST	n-1		

$$H_0 : \beta_0 = \beta_1 = 0$$

$$H_1 : \beta_0 \neq 0 \text{ or } \beta_1 \neq 0$$

$$\text{Test Statistic: } F_0 = \frac{SSR}{SSE/n-2}$$

If $F_0 > F_{1,n-2,\alpha}$, then H_0 is rejected.